

Solve Multiplication and Division Equations

Lesson 7-3

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$x + 2 = 7$$

balanced with =

*Multiplication and division are inverses: $3 \times 7 = 21$,
so $21 \div 3 = 7$*

Inverse operation

Two math actions that undo each other, like $\times$ and $\div$.



$$3 \times 2 = 6$$

$3 \times 7 = 21$ means 3 groups of 7 equals 21

Multiply

To add equal groups again and again.



$$6 \div 3 = 2$$

*$36 \div 4 = 9$ means 36 split into 4 equal groups gives
9 per group*

Divide

To split into equal groups.

$$x + 2 = 7$$

balanced with =

$3x = 21 \rightarrow$ divide both sides by 3 $\rightarrow x = 7$ (x is alone)

Isolate

To get the letter by itself on one side.


$$4x$$

coefficient = 4

In $6x = 18$, the coefficient is 6; divide both sides by 6 to get $x = 3$.

Coefficient

The number multiplied by a variable.

$$x + 2 = 7$$

balanced with =

$x = 3$ is the solution of $6x = 18$ because $6 \times 3 = 18$.

Solution

The value of the variable that makes the equation true.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can solve one-step multiplication and division equations using inverse operations.

1 Notice

Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence.

2 Key idea

I can solve one-step multiplication and division equations using inverse operations.

3 Words I'll use

Inverse operation

Multiply

Divide

Isolate

Coefficient

Solution



Think About It

- What does the 3 represent in the equation?
- What does x represent?
- What operation connects 3 and x?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Solve: $6x = 42$

- A. $x = 7$
- B. $x = 48$
- C. $x = 36$
- D. $x = 252$

Step 1 Divide both sides by 6: $x = 42 \div 6 = 7$.

Step 2 Check: $6 \times 7 = 42$ ✓



Answer: A. $x = 7$

 We do — together

Solve this one with your class using the same steps.

Problem: Solve: $n / 3 = 8$

A. $n = 24$

B. $n = 11$

C. $n = 5$

D. $n = 3$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Solve: $9m = 63$

A. $m = 7$

B. $m = 54$

C. $m = 72$

D. $m = 567$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Detective Park has 3 identical boxes holding 21 pieces of evidence total, written as $3x = 21$. What does each part of the equation mean, and how is solving this different from solving $x + 3 = 21$?

Level 1 support

Start here: In $3x = 21$ the variable is multiplied, so I undo it with division.

Need a hint?

Sentence starters (optional)

The 3 means ____ and x means ____.

El 3 significa ____ y x significa ____.

Solving $3x = 21$ is different from $x + 3 = 21$ because ____.

Resolver $3x = 21$ es diferente de $x + 3 = 21$ porque ____.

Word bank: inverse operation multiply divide isolate

Listen for: Students read $3x$ as 3 groups of x , recognize multiplication, and contrast it with addition, knowing they will divide (not subtract) to solve.

Level 2

Predict: would x in $3x = 21$ be larger or smaller than in $5x = 21$? Justify before solving.

- x would be ____ in $5x = 21$ because ____.
- More boxes means each box has ____.

EXPLORE

How did you undo the multiplication in $3x = 21$ (or the division in $x / 4 = 9$) to isolate the variable?

Level 1 support

Start here: I use the inverse operation: division undoes multiplication.



Need a hint?

Sentence starters (optional)

I divided / multiplied both sides by _____, so the answer is _____.

Dividí / multipliqué ambos lados por _____, así que la respuesta es _____.

The inverse of _____ is _____, so I _____.

El inverso de _____ es _____, así que _____.

Word bank: inverse operation multiply divide isolate

Listen for: Students divide both sides by 3 to get $x = 7$ (or multiply by 4 to get $x = 36$) and name division/multiplication as inverse operations.

Level 2

Why does dividing only one side of $3x = 21$ give the wrong answer? Compare doing it to one side versus both sides.

- Dividing only one side is wrong because _____.
- To keep the equation true I must _____.

CONNECT

When would you solve a multiplication or division equation in real life?

Level 1 support

Start here: These equations find an unknown when items are split into or made from equal groups.



Need a hint?

Sentence starters (optional)

I would use this kind of equation to find ____.

Usaría este tipo de ecuación para hallar ____.

I would ____ both sides because ____.

____ ambos lados porque ____.

Word bank: **inverse operation** **multiply** **divide** **isolate**

Listen for: Students name a real context (equal price per item, sharing equally, total from groups) and choose the correct inverse operation to isolate the variable.

Level 2

Critique: 'To solve $x / 5 = 12$ you divide both sides by 5.' Explain the error and the correct inverse operation.

- This is wrong because the inverse of division is ____.
- I should ____ both sides by 5, which gives ____.

Write About the Math

The Writing Revolution

I can explain my steps using the words inverse operation, multiply, divide, and isolate.

1. Kernel Sentence subject + verb

Model: Solution is the value of the variable that makes the equation true.

Solución es el valor de la variable que hace verdadera la ecuación.

Write a kernel sentence about solution. Use a subject and a verb.

Escribe una oración base sobre solución. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Solution matters in math

Solución importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Solution matters in math because ____.

Solución importa en matemáticas porque ____.

but
pero

Solution matters in math, but ____.

Solución importa en matemáticas, pero ____.

so
entonces

Solution matters in math, so ____.

Solución importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about solution.
Di un hecho verdadero sobre solution.

Solution ____.

Question
Pregunta

Ask a question about solution.
Haz una pregunta sobre solution.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about solution.
Muestra entusiasmo sobre solution.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with solution.
Dile a un compañero qué hacer con solution.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The 3 means ____ **and x means** ____ .

El 3 significa ____ *y x significa* ____ .

Solving $3x = 21$ is different from $x + 3 = 21$ because ____ .

Resolver $3x = 21$ es diferente de $x + 3 = 21$ porque ____ .

I divided / multiplied both sides by ____ , **so the answer is** ____ .

Dividí / multipliqué ambos lados por ____ , *así que la respuesta es* ____ .

Try It

Solve on your own. Check the answer key when you are done.

1. Solve: $8n = 2$.

A. $n = 0.25$

B. $n = 16$

C. $n = 4$

D. $n = 10$

Show your work:

2. Five friends split a bill equally and each pays \$12. Which equation finds the total t ?

A. $t/5 = 12$

B. $5t = 12$

C. $t - 5 = 12$

D. $t + 5 = 12$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bakery packed muffins into boxes of 8. They made 96 muffins total. Write an equation, solve it, and explain why your answer is reasonable. Then write a different situation that uses division and has the same answer.

Sentence starter: Equation: _____. Solving: _____. The answer _____ is reasonable because _____. A division situation with the same answer: _____.

Show your work:

Reflect — Exit Ticket

Solve: $w / 8 = 6$

- A. $w = 48$
- B. $w = 14$
- C. $w = 0.75$
- D. $w = 2$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $n = 24$ — *Multiply both sides by 3: $n = 8 \times 3 = 24$. Check: $24 / 3 = 8$ ✓*
2. **You Try (notes):** A. $m = 7$ — *Divide both sides by 9: $m = 63 \div 9 = 7$. Check: $9 \times 7 = 63$ ✓*
3. **Try It 1:** A. $n = 0.25$ — *$n = 2 \div 8 = 0.25$.*
4. **Try It 2:** A. $t/5 = 12$ — *Total divided by 5 equals 12, so $t = 60$.*
5. **Exit Ticket:** A. $w = 48$ — *Multiply both sides by 8: $w = 6 \times 8 = 48$. Check: $48 / 8 = 6$ ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Solution is the value of the variable that makes the equation true.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).